Prosper Magara – Curriculum Vitae

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Contact

Cape Town, South Africa. Available to relocate to KwaZulu-Natal and to travel up to 50% of the time.

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- Phone: +27 63 065 2215
- Google Scholar: scholar.google.com/citations?user=zAdTzMwAAAAJ
- GitHub: github.com/Prosper02
- Portfolio: prosper02.github.io/prosper1

Professional summary

Biomedical and electrical engineer with hands-on medical equipment field experience since 2020 and a research focus on robot-assisted laparoscopic surgery. I have planned, installed, calibrated and serviced X-ray machines, dialysis units, autoclaves and chemistry analysers in hospitals across Zimbabwe, and supported an MRI installation and a water-treatment plant at Parirenyatwa Group of Hospitals.

Since 2024 I have worked with surgeons at Groote Schuur Hospital as a Google PhD Fellow at the University of Cape Town, building telementoring systems for laparoscopic and da Vinci robot-assisted surgery. That work covers the whole chain a field service engineer touches: camera and video hardware, network latency, software updates, remote diagnostics and operating-theatre protocol, plus the relationships with surgeons and theatre staff that keep systems in use.

I troubleshoot complex electro-mechanical and software systems methodically, keep clean service records, and communicate clearly with clinical and technical teams. Available to relocate to KwaZulu-Natal and to travel up to 50% of the time.

Core competencies

Area	What I do
Medical equipment field service	Pre-site planning, delivery, installation and commissioning; calibration; preventive maintenance scheduling; fault diagnosis and repair of X-ray, dialysis, autoclave, theatre and laboratory equipment; spare parts and service reporting
Surgical robotics and laparoscopy	Hands-on experience with the da Vinci surgical robot at Groote Schuur Hospital; LapSim virtual-reality simulator and laparoscopic box trainers; laparoscopic camera and video chains; operating-theatre protocol
Diagnostics and software	Troubleshooting electro-mechanical and embedded systems (PIC, STM32, ATmega328); remote diagnostics and latency measurement tooling; Windows and Linux; Python, C#, JavaScript; networking (Wi-Fi, GSM, 5G, WebRTC)
Customers and administration	Working relationships with surgeons, nurses and hospital technical staff; service reports, inventories and maintenance records; technical writing and presentations; supervising junior engineers

Professional experience

PhD Researcher and Google PhD Fellow · Division of Biomedical Engineering, University of Cape Town · January 2024 to present

- Work with surgical teams at Groote Schuur Hospital on telementoring for laparoscopic and da Vinci robot-assisted surgery, with hands-on time on the da Vinci system, the LapSim simulator and laparoscopic box trainers.
- Built a real-time telementoring platform (dual-camera WebRTC streaming, telestration, two-way audio) and a mentee-side theatre application with real-time face blurring.
- Designed and built a glass-to-glass latency measurement rig (ATmega328 microcontroller, light sensors, desktop analysis software) for diagnosing video delay across endoscope, capture and network chains.

- Lead author of a systematic review of AI bleeding monitoring in conventional and robot-assisted laparoscopic surgery (Journal of Robotic Surgery, 2026) and of a latency-aware evaluation of bleeding detection algorithms (Springer, 2026).
- Ran user-centred design work with surgeons to specify system requirements, published at the 2024 IEEE Globecom Workshops.

Medical Device Engineer · Med-Elect Technologies, Zimbabwe · 2022 to 2023

- Planned and carried out installations of biomedical equipment, including X-ray machines, dialysis machines, autoclaves, theatre beds and chemistry analysers, at hospitals and clinics.
- Serviced, calibrated and repaired equipment on site; scheduled preventive maintenance and kept service records for the installed base.
- Liaised with hospital management and clinical staff on site readiness, handover and user training.
- Designed and installed solar backup power for medical equipment at sites with unreliable grid supply.

Biomedical Engineering Attaché · Parirenyatwa Group of Hospitals, Harare · 2022, six months

- Equipment inventory, installation and maintenance across general surgery, paediatrics, oncology, haematology, orthopaedics and radiography.
- Took part in the installation of a new MRI scanner and a hospital water-treatment plant.

Biomedical and IoT Engineer · SolidarMed, Zimbabwe · 2020 to 2021

- Developed IoT health devices, including a smart pill bottle, and implemented telemedicine tooling (REDCap) under the electronic health project.
- Monitoring and evaluation and technical report writing; supervised three computer science students and one electronic engineering student on attachment.

Freelance Solar Systems Engineer · Zimbabwe · 2018 to 2023

- Designed, installed and maintained off-grid and hybrid solar PV systems for homes, farms and clinics, from load sizing to inverter and battery commissioning.

Education

Years	Qualification	Institution
2024 to present	PhD, Biomedical Engineering. Telementoring and telesurgery with augmented reality and haptics over 5G networks	University of Cape Town

Years	Qualification	Institution
2021 to 2022	MSc, Biomedical Engineering (biomedical devices and imaging), distinction. Thesis: a smart health station for swab collection and patient data entry	University of Zimbabwe
2015 to 2019	BSc (Hons), Electrical Engineering, upper second class. Final-year projects: a smart egg incubator and an LME49710-based audio amplifier	University of Zimbabwe

Awards and fellowships

Year	Award
2026	PhD First Prize, Best Presenter Award, HUB Biomedical Engineering Fusion event, University of Cape Town
2024	Google PhD Fellowship, for research combining AI and augmented reality in telementoring systems for laparoscopic surgery
2019	Best Innovative Project, University of Zimbabwe Research Week, for the smart egg incubator
	Second Creative SDGs Idea, CBZ Bank Hacking for SDGs post COVID-19 hackathon, Zimbabwe

Publications

1. Magara P, Sivarasu S, Ras L, Biyabani A, Blocker A, Shabangu M, Malila B. AI-based automated bleeding monitoring in conventional and robot-assisted laparoscopic surgery: a systematic review. *Journal of Robotic Surgery* 2026;20(1):522. doi:10.1007/s11701-026-03469-4
2. Magara P, Sivarasu S, Ras L, Biyabani A, Malila B. Latency-aware evaluation of bleeding monitoring algorithms for real-time surgical applications. *Nordic Conference on Digital Health and Wireless Solutions*, Springer, 2026, pp. 51–69. doi:10.1007/978-3-032-28819-6_4
3. Blocker A, Magara P, Biyabani A, Datay MI, Mwangama J, Malila B. Design of a solar-based virtual clinic for austere environments. *2025 IEEE AFRICON*, pp. 1–8. IEEE Xplore

4. Magara P, Ras L, Biyabani A, Malila B. User-centred design of a tele-mentoring system for laparoscopic surgery in austere environments. *2024 IEEE Globecom Workshops (GC Wkshps)*, pp. 1–7. IEEE Xplore
5. Magara P, Saruchera G, Mushiri T, Noorbhai H. Development of a semi-automatic, isolating and portable health-station for swab collection and patient data entry. University of Zimbabwe, 2022.

Additional information

- Languages: English and Shona, both native.
- Driver's licence: valid licence with a clean driving record.
- Research groups: Telemedicine and Connected Care (2024 to 2027) and UCT MedTech (2025 to 2027), University of Cape Town.
- Research travel: Rwanda, Italy and Finland.
- Referees: available on request.